



SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis in Computational Science and Engineering

**Fast Multipole Boundary Element Method
for Finite Periodic Structures**

Abhijeet Abhay Sutar





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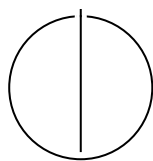
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Master's Thesis in Computational Science and Engineering

**Fast Multipole Boundary Element Method
for Finite Periodic Structures**

Titel der Abschlussarbeit

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Submission Date:	Submission date



I confirm that this master's thesis is my own work and I have documented all sources and material used.

Munich, Submission date

Abhijeet Abhay Sutar

Acknowledgments

Abstract

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1 Introduction

1.1 Section

Citation test [Lam94].

Acronyms must be added in `main.tex` and are referenced using macros. The first occurrence is automatically replaced with the long version of the acronym, while all subsequent usages use the abbreviation.

E.g. `\ac{TUM}`, `\ac{TUM}` $\Rightarrow$ Technical University of Munich (TUM), TUM

For more details, see the documentation of the acronym package¹.

1.1.1 Subsection

See Table 1.1, Figure 1.1, Figure 1.2, Figure 1.3.

Table 1.1: An example for a simple table.

A	B	C	D
1	2	1	2
2	3	2	3

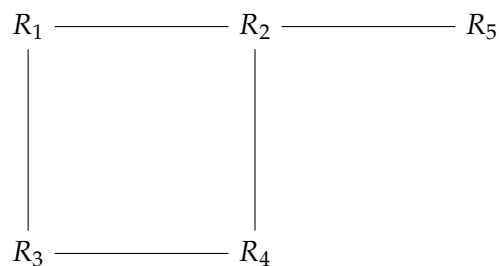


Figure 1.1: An example for a simple drawing.

¹<https://ctan.org/pkg/acronym>

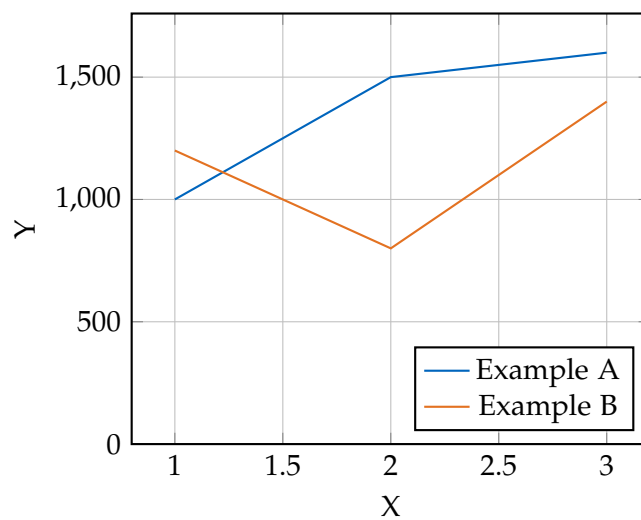


Figure 1.2: An example for a simple plot.

```
SELECT * FROM tbl WHERE tbl.str = "str"
```

Figure 1.3: An example for a source code listing.

2 Terms and Definitions

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3 Mathematical Formulation of BEM

3.1 Acoustic Wave Equation

The propagation of acoustic waves in a homogeneous, inviscid fluid medium is governed by the linear wave equation for the acoustic pressure $p(\mathbf{x}, t)$:

$$\nabla^2 p(\mathbf{x}, t) - \frac{1}{c^2} \frac{\partial^2 p(\mathbf{x}, t)}{\partial t^2} = 0 \quad (3.1)$$

where $\mathbf{x}$ is the position vector, t is time, and c is the speed of sound in the medium.

To derive the Helmholtz equation, we consider a time-harmonic acoustic field. We assume the pressure can be represented as a complex function oscillating with angular frequency ω :

$$p(\mathbf{x}, t) = P(\mathbf{x})e^{-i\omega t} \quad (3.2)$$

where $P(\mathbf{x})$ is the spatial component of the pressure (the phasor) and $i = \sqrt{-1}$ is the imaginary unit.

Substituting Equation (3.2) into the wave equation (3.1):

$$\nabla^2 (P(\mathbf{x})e^{-i\omega t}) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} (P(\mathbf{x})e^{-i\omega t}) = 0 \quad (3.3)$$

Calculating the time derivatives:

$$\frac{\partial}{\partial t} (P(\mathbf{x})e^{-i\omega t}) = -i\omega P(\mathbf{x})e^{-i\omega t} \quad (3.4)$$

$$\frac{\partial^2}{\partial t^2} (P(\mathbf{x})e^{-i\omega t}) = (-i\omega)^2 P(\mathbf{x})e^{-i\omega t} = -\omega^2 P(\mathbf{x})e^{-i\omega t} \quad (3.5)$$

Substituting this back into the wave equation:

$$e^{-i\omega t} \nabla^2 P(\mathbf{x}) - \frac{1}{c^2} (-\omega^2 P(\mathbf{x})e^{-i\omega t}) = 0 \quad (3.6)$$

Dividing by the common time factor $e^{-i\omega t}$ (which is non-zero) and rearranging the terms:

$$\nabla^2 P(\mathbf{x}) + \frac{\omega^2}{c^2} P(\mathbf{x}) = 0 \quad (3.7)$$

Introducing the wavenumber $k = \omega/c$, we arrive at the homogeneous Helmholtz equation:

$$\nabla^2 P(\mathbf{x}) + k^2 P(\mathbf{x}) = 0 \quad (3.8)$$

3.2 Boundary Element Method

3.2.1 Boundary Integral Formulation

The Boundary Element Method (BEM) solves the Helmholtz equation by transforming it into an integral equation defined on the boundary of the domain. This transformation is typically achieved using Green's Second Identity.

Let Ω be the domain containing the fluid and Γ be its boundary. For two scalar fields P and G in Ω , Green's Second Identity states [Wu00]:

$$\int_{\Omega} (P \nabla^2 G - G \nabla^2 P) d\Omega = \int_{\Gamma} \left(P \frac{\partial G}{\partial n} - G \frac{\partial P}{\partial n} \right) d\Gamma \quad (3.9)$$

where n is the outward unit normal vector to the boundary.

We choose $P(\mathbf{x})$ to be the acoustic pressure, satisfying the Helmholtz equation derived in Equation (3.8):

$$\nabla^2 P(\mathbf{x}) + k^2 P(\mathbf{x}) = 0 \quad (3.10)$$

We choose $G(\mathbf{x}, \mathbf{y})$ to be the fundamental solution (Green's function) of the Helmholtz equation, which satisfies the inhomogeneous equation with a point source at $\mathbf{y}$:

$$\nabla^2 G(\mathbf{x}, \mathbf{y}) + k^2 G(\mathbf{x}, \mathbf{y}) = -\delta(\mathbf{x} - \mathbf{y}) \quad (3.11)$$

where δ is the Dirac delta function. For 3D acoustics, the free-space Green's function satisfying the Sommerfeld radiation condition is given by [Kir98]:

$$G(\mathbf{x}, \mathbf{y}) = \frac{e^{ikr}}{4\pi r} \quad \text{with} \quad r = |\mathbf{x} - \mathbf{y}| \quad (3.12)$$

Substituting these into Green's Second Identity for a source point $\mathbf{y}$ inside the domain Ω :

$$P(\mathbf{y}) = \int_{\Gamma} \left(G(\mathbf{x}, \mathbf{y}) \frac{\partial P}{\partial n}(\mathbf{x}) - P(\mathbf{x}) \frac{\partial G}{\partial n}(\mathbf{x}, \mathbf{y}) \right) d\Gamma(\mathbf{x}) \quad (3.13)$$

As the source point $\mathbf{y}$ is moved to the boundary Γ , the integrals become singular. Evaluating the singularity using a limiting process yields the Boundary Integral Equation (BIE) [Mar08]:

$$C(\mathbf{y})P(\mathbf{y}) = \int_{\Gamma} \left(G(\mathbf{x}, \mathbf{y}) \frac{\partial P}{\partial n}(\mathbf{x}) - P(\mathbf{x}) \frac{\partial G}{\partial n}(\mathbf{x}, \mathbf{y}) \right) d\Gamma(\mathbf{x}) \quad (3.14)$$

where $C(\mathbf{y})$ is the geometric coefficient (solid angle). For a smooth boundary, $C(\mathbf{y}) = 0.5$.

3.3 Discretization

To solve the BIE numerically, the boundary Γ is discretized into N boundary elements. The pressure P and its normal derivative $q = \partial P / \partial n$ are approximated using shape functions.

Applying the collocation method at the nodes leads to a system of linear algebraic equations:

$$\mathbf{H}\mathbf{P} = \mathbf{G}\mathbf{Q} \quad (3.15)$$

where $\mathbf{H}$ and $\mathbf{G}$ are the dense, nonsymmetric influence matrices resulting from the integration of the fundamental solution and its derivative, and $\mathbf{P}$ and $\mathbf{Q}$ are vectors containing the nodal values of pressure and normal velocity, respectively.

3.4 Burton-Miller Formulation

The Boundary Element Method (BEM) offers a highly elegant, mathematically rigorous, and computationally efficient alternative by reformulating the volumetric partial differential equation into a Boundary Integral Equation (BIE) mapped strictly over the surface of the scattering or radiating body. This approach intrinsically satisfies the Sommerfeld radiation condition at infinity, thereby reducing the dimensionality of the problem by one—transforming a three-dimensional volumetric domain into a two-dimensional surface mesh, or a two-dimensional problem into a one-dimensional contour [Lam94; Wu00; Mar08; Kir98].

However, as [BM71] showed, the standard Boundary Element Method (BEM) approach for exterior problems, gives rise to non-existence of unique solutions at wavenumbers coinciding with the resonances of the corresponding interior problem.

At these critical frequencies, also sometime referred to as *fictitious or irregular frequencies*, the matrix system resulting from the discretization of the BIE becomes ill-

conditioned, leading to numerical instability and a spike in the error of the computed solution.

It must be noted that the original exterior problem does have a unique solution at these frequencies and that the connection to the interior problem has no physical significance. The non-uniqueness arises solely from the formulation of the problem in terms of a Boundary Integral Equation (BIE).[BM71]

4 The Fast Multipole Method Framework

The BEM provide an elegant approach to modelling acoustic wave propagation. It allows for the modelling of finite structures in an infinite exterior domain without needing any special tricks for the boundaries. However, the required mathematical transformation of the problem results in a dense full matrix, that is both complex and non-Hermitian [CL07]. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

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5 Software Architecture and HPC Optimizations

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6 Performance Analysis and Validation

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7 Conclusion and Future Directions

- Currently Burton - Miller is highly optimized for single dof per cell and for perfectly flat panels. That is, $FE_Degree = 0$ and $Mapping_Degree = 1$. To support higher degrees, we need to update the hypersingular calculation in the near field operator and the main assemble system method.
- There is some basic parallelization in the code for computing the translation matrices and for the matrix-vector multiplication done with OpenMP pragmas. But, Deal.ii has its native parallelization support using its native datatypes. The FMM also lends itself to parallelization where each node of the FMM tree may reside in an individual process and maintain its own near field matrix, with only the far-off node dipoles having to be passed around.
- The current implementation of the computation of the external field uses direct computations of the green's function. This can be upgraded to use the FMM approximation in the future, which will be faster for large problems.
- The current implementation is strictly for scattering off of sound hard objects. The next step would be to implement the option to set an arbitrary impedance boundary condition.
- The current implementation is strictly for a stationary mesh.

Abbreviations

TUM Technical University of Munich

BEM Boundary Element Method

BIE Boundary Integral Equation

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Bibliography

- [BM71] A. J. Burton and G. F. Miller. “The Application of Integral Equation Methods to the Numerical Solution of Some Exterior Boundary-Value Problems.” In: *Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences* 323.1553 (June 8, 1971), pp. 201–210. ISSN: 0080-4630, 2053-9169. DOI: 10.1098/rspa.1971.0097.
- [CL07] S. Chandler-Wilde and S. Langdon. “Boundary Element Methods for Acoustics.” In: (July 18, 2007).
- [Kir98] S. Kirkup. *The Boundary Element Method in Acoustics: A Development in Fortran*. Integral Equation Methods in Engineering 1. Hebden Bridge: Integrated Sound Software, 1998. 136 pp. ISBN: 978-0-9534031-0-3.
- [Lam94] L. Lamport. *LaTeX : A Documentation Preparation System User’s Guide and Reference Manual*. Addison-Wesley Professional, 1994.
- [Mar08] S. Marburg. “Boundary Element Method for Time-Harmonic Acoustic Problems.” In: *Computational Acoustics of Noise Propagation in Fluids*. Ed. by S. Marburg and B. Nolte. Berlin, Heidelberg: Springer, 2008.
- [Wu00] T. Wu. *Boundary Element Acoustics: Fundamentals and Computer Codes*. Southampton, UK: WIT Press, 2000.